



ENGS 76 - Machine Engineering

Fall 2025

Bridge Building Robot

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Problem Definition

0.1 Problem Statement

This project is the pinnacle project for a Dartmouth Mechanical Engineering student. Over the course of the project, a team is tasked with designing, fabricating, and operating a joystick-operated robot capable of building a bridge. The robot¹ must be able to build this bridge given a variety of materials at its disposal on a controlled playing surface. The given materials, including I-beams, plates, and a ramp, are not prohibitively heavy but present an exciting design challenge when considering how to effectively manipulate each with one crane. The project culminates in a competition in which students face off against each other on opposite sides of the playing surface's 34-inch trench, shown below. Teams operate the robot virtually via remote control and must complete the task safely, reliably, and efficiently in order to maximize points scored. The rules also constrain the machine to strict dimensional, material, and fabrication requirements.



Figure 1: The playing field on competition day

¹see Figures 1.5, 1.3 for our fabricated robot and SOLIDWORKS assembly view

0.2 Specifications

The following list is a combination of specifications, both dictated by the rulebook and developed by our team as goals to augment our team's success in the competition:

Required Specifications

- **Working Environment Requirements** — The robot must operate within the arena consisting of two aligned trenches each 12" wide and 5" deep. The robot must be contained in a 12" x 12" x 48" footprint.
- **Functionality** — The machine must be able to move through the arena to accurately grasp materials securely and transport them with precision. The machine cannot harm construction workers placed on the board and must remove building materials from the bridge so a car can drive past.
- **Timeframe** — 8 minute time limit to deploy and construct the bridge without damaging arena components.
- **Materials** — materials must be a part of the provided list of approved materials.
- **Fabrication Techniques** — there must be 1 milled, 1 turned, 1 3d printed and 1 laser cut part, included in 4 total unique milled/turned parts.

Our Specifications/Design Requirements

- **Chassis** — Weight minimization, target weight: under 5 lbs
- **Gripper** — Able to lift two I-beams at once. Unique, simplistic, gripper geometry optimized to interact with thin plates, boxy ramp surfaces and cylindrical cones, fail safe to protect workers. Capacity to pick up stickers (wood pieces for spacing aluminum plates). Sufficient clamping force to move I-beams accross the full range of motion without slippage.
- **Controller** — Optimized channels and settings to ensure intuitive bridge building.
- **Subsystem Integration** — All systems are modular to allow for quick and effective repairs. Cable management avoids snagging, tangling, and pulling of cords.

Design Considerations

To solve the bridge-building challenge, our team evaluated several design concepts before selecting a crane-style machine capable of reaching across the trench and precisely manipulating all required game pieces. We briefly considered alternatives such as a wheeled mobile robot, a fixed arm with limited reach, and a telescoping linear actuator system. These concepts were ultimately rejected because they offered less range of motion, introduced greater stability concerns, or required fabrication techniques that exceeded the project's constraints. We selected a rotating crane design because it provided the greatest operational flexibility with the simplest mechanical interface. The system consisted of five major modules:

- **Base Rotation Module** — The final form of this module was a lazy-Susan style turntable built with aluminum rollers acting as bearings, allowing smooth 360° rotation of the upper structure. The top of this base rotation module was designed to cleanly house a glue joint that anchored the crane tower.

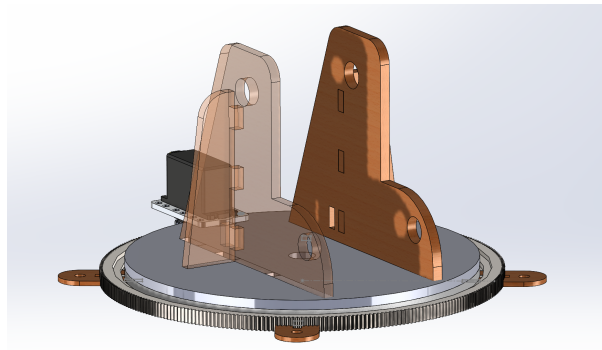


Figure 1.1: Final Rotation Assembly With Tower Interface

- **Lateral Translation Cart (Figure A.4)** — A pulley actuated carriage that enabled the gripper to move horizontally across the span, ensuring we could accurately position roadbed plates, I-beams, and cones.
- **Rotational Gripper Assembly (Figure A.1)** — A custom end effector (3D Printed D-Shaft) with a pivot joint that allowed the gripper to rotate into optimal orientations for lifting and placing each type of game piece.
- **Gripper Jaw Assembly (Figure A.2)** — A gear and rack gripper with custom grasping claws, designed to optimally grasp objects in the playing field. The gripper was designed to sit under the object it was holding—this solution prevented the gripper from relying on a servo's torque for clamping force.
- **Vertical Motion** — This module was responsible for moving the crane arm up and down, and therefore enabling the raising and lowering of the gripper. At first, inspired by the non-back-drivable nature of a worm-gear, we designed a gearbox that utilized such a gear to spin a pulley (see Figure 5.2). This design proved effective but inelegant. It was extremely loud given the high speed of operation

required for sufficient angular output. Furthermore, it took up a significant amount of valuable space in the base that was better used for structural elements. Thus, in our second design push (more on that later), we designed a system inspired by industrial construction vehicles, that used a threaded linear actuator to raise and lower the arm (see Figure A.3).

Our design process followed a two-phase approach. First, we intentionally fabricated a fully working, but non-optimized, prototype to explore the problem space, test mechanical ideas, and expose failure modes early. This rapid-learning prototype served the same role as scaled-down prototypes in industry: enabling maximum insight with minimal time and resources. Guided by what we learned from the prototype, our team designed and fabricated a refined, optimized crane-style bot for competition day. This final design incorporated improved mechanisms, tighter tolerances, and enhanced reliability while maintaining the successful aspects of the initial prototype.

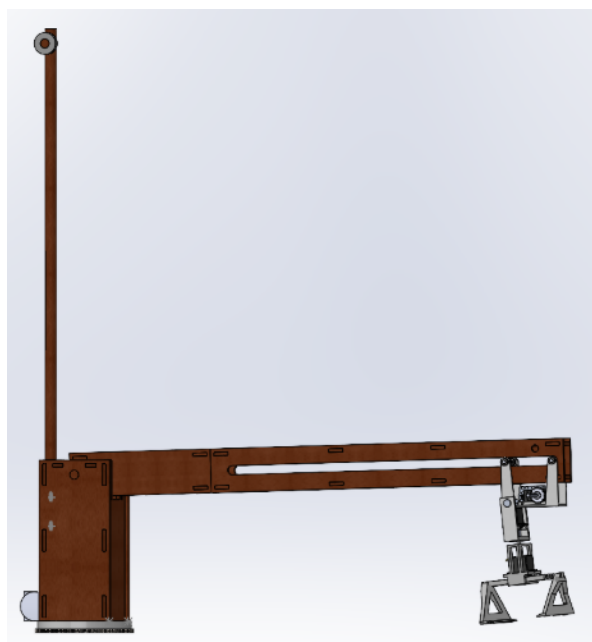


Figure 1.2: First Full Design

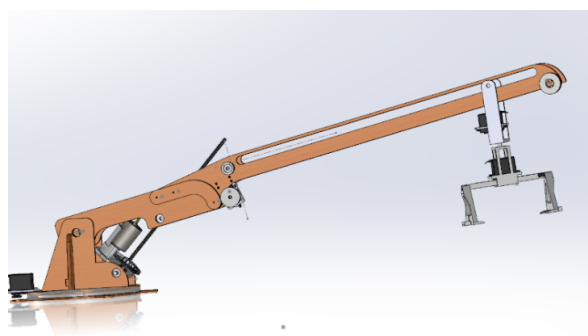


Figure 1.3: Final Design

Figure 1.4: A side-by-side of design evolution

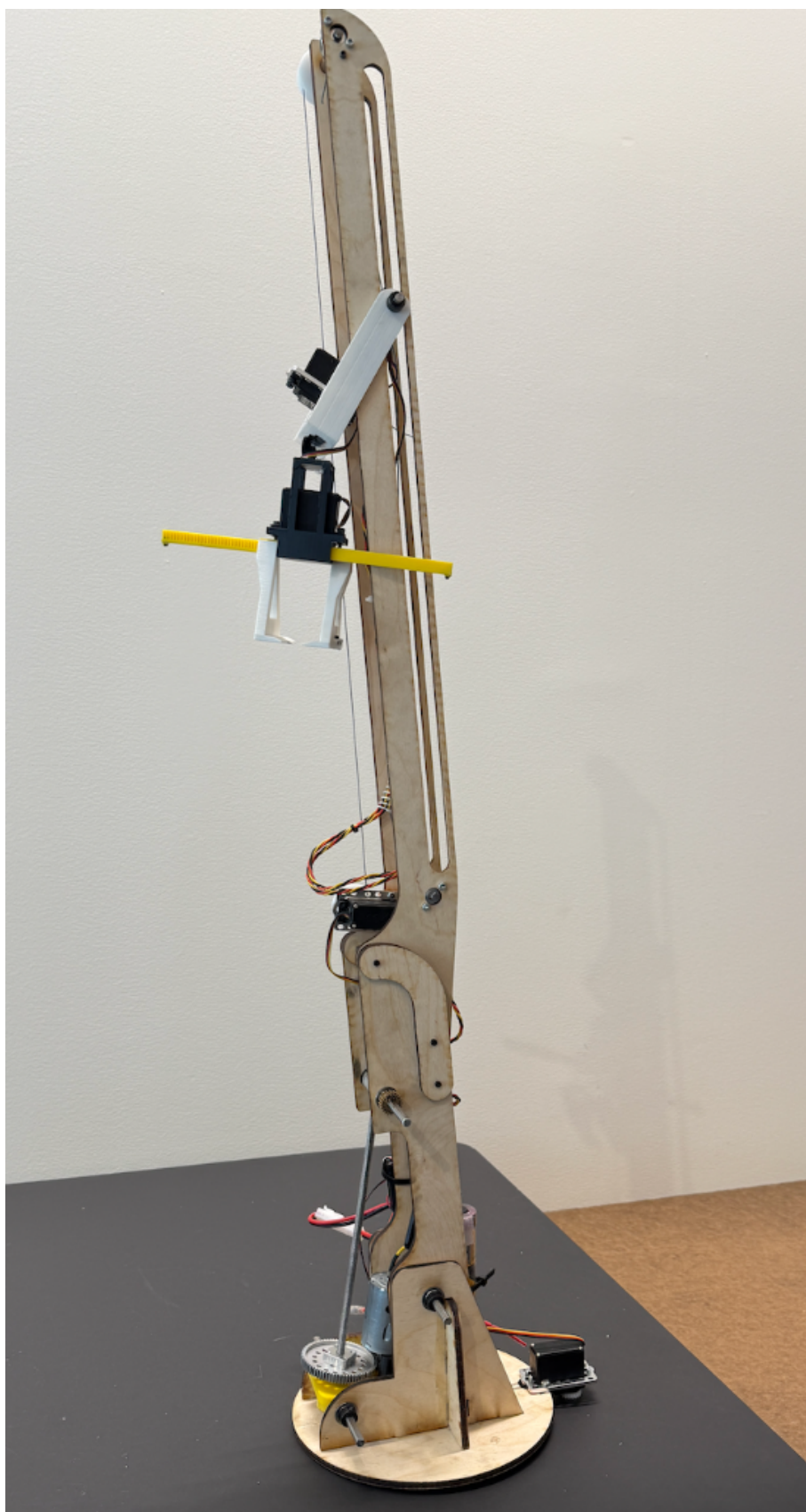


Figure 1.5: The final product

Analysis

The key analysis driving our design process was how to build the most efficient and reliable robot possible. Instead of settling for the first “workable” concept, we used structured decision frameworks, such as FAARBARC, to evaluate alternatives and justify our choices. We selected a crane architecture because it minimized control system complexity: unlike channel access designs, the crane allowed full reach without requiring multiple coordinated mechanisms. For vertical motion, we chose a threaded linear actuator rather than a pulley system because it offered higher reliability, faster response, reduced part count, and lower overall weight, while also being far easier to maintain. We reduced unnecessary material in the chassis to streamline the robot and improve maneuverability. Also, by using Euler buckling criteria, we settled on eliminating front and back walls from our main tower design to minimize weight and simplify the installation process of the bot. We also conducted a series of test cuts on the laser cutter to determine the best tolerances necessary to produce clearance or interference fits. Finally, we iterated on the gripper geometry to ensure that every game piece could be picked up efficiently, securely, and with repeatable control.

Apart from those mentioned, we conducted a series of mechanical and structural analyses of our modules over the series of the project:

- Speed-torque analyses for calculated loads on the threaded linear actuator
- Speed-torque analyses for the loads on the pulley system
- Shear and moment analyses to determine the minimum cross-sectional thickness and moment of inertia of the crane arm
- Axial and bending stress analyses to determine gripper tooth geometry

A collection of these specific calculations are provided in the **appendix**.

Design details

3.1 Final Bill of Materials

The following is our final updated bill of materials.

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	003 The Boring Plate – Base	Plate to secure the lazy susan, Laser Cut Plywood	1
2	002 Bearing Retainer	Holds rotational bearings on lazy susan, 3D printed PLA	1
3	001 Tooth Reciever	Crane mounting plate for rotational servo and crane, Laser Cut Plywood	1
4	OSNEW076-EL-P006	Continuous rotation servo, from kit	4
5	076-ME-P030 Servo Mounting Plate	Servo mounting plate, from kit	3
6	076-ME-P010	Servo spur gear, 20 tooth, from kit	1
7	076-ME-P007-1	Hub gear, 72 tooth, 0.5in bore, from kit	1
8	001 Side Walls	Side wall of tower, laser cut plywood	2
9	002 Tower Fins	Base of crane side supports, laser cut plywood	2
10	634094	D-shaft, 4in	1
11	004 Ring Gear	Lazy Susan big gear, 3D printed PLA	1
12	94210A003	Medium-strength threaded rod	1
13	076-ME-P025	Flanged ball bearings, 0.25in bore, OD = 0.625in, from kit	14

Table 3.1: Bill of Materials (Items 1–13)

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
14	001 Rod Coupler	Counterpart to threaded rod allowing raising and lowering of crane, machined (lathe + mill) aluminum	1
15	076-ME-P006	Hub gear, 72 tooth, 0.5in bore, from kit	1
16	076-ME-P007-1	Hub mount, 0.25in bore, from kit	1
17	076-EL-P003	Tamiya 540 stock motor, from kit	1
18	076-ME-P015	Compression spring, $K = 6.16$, $L = 1$ in, from kit	2
19	005 Motor Cradle	Secures the motor in place, 3D printed PLA	1
20	006 Laser Screw Tab	Stabilizes mounting holes on motor cradle, laser cut plywood	1
21	002 Arm Wall	Crane arm with servo mounting holes, laser cut plywood	1
22	001 Tower Interface	Shorter crane arm attaching arm walls to side walls, laser cut plywood	2
23	002 Arm Wall	Crane arm with hub mount mounting holes, laser cut plywood	1
24	003 Pressure Slat	Reinforces connection point between tower interface and arm wall	2
25	076-ME-P026	Standoffs, 1.5in, plates to allow for spacing and support between arm walls, from kit	4
26	5506076	Socket cap screws, $4-40 \times 3/4$ "	4
27	076-ME-P030	Servo mounting plate, from kit	3
28	002 Servo Mounting Spacer	Separates servo mounting plate from arm wall for alignment, 3D printed PLA	1
29	Pulley 1.5 Diameter	Pulley wheel for lateral pulley system, machined on lathe (plastic)	1
30	076-ME-P031S	Servo shaft coupler, 0.25in bore, from kit	2
31	004 High Pulley	Pulley wheel for lateral pulley system, 1in, 3D printed PLA	1

Table 3.2: Bill of Materials (Items 14–31)

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
31	004 High Pulley	Pulley wheel for lateral pulley system, 1in, 3D printed PLA	1
32	001 New Servo Pulley	Driver pulley wheel attached to servo, groove lined with rubber band, 3D printed PLA	1
33	Nov6-Gripper Box	Gripper cage securing servo for gripper arm movement, 3D printed PLA	1
34	001 Cart and Rotation Mechanism	Connects gripper sub-assembly to arm and lateral pulley system, contains servo for gripper rotation, 3D printed PLA	1
35	32000515	0.25in aluminum rod, 1in, cut to fit in lazy susan slots	8
36	D Shaft 4in length	Connects the sides of bot arms	5
37	Spur Gear 16 tooth 0.25in Bore	From kit	2
38	73125080	Cable ties, small nylon, 4in long	4
39	327 Yds-80LB	Braided fishing line, 150lb	1
40	076-ME-P013	Shaft collar, 0.25in bore	12
41	87920708	Lock washers, 4-40	4
42	52592706	6-32 hex lock nuts	4
43	Nov12-rack	Gripper rack, 3D printed PLA	2
44	Gripper Blades	Machined / ground down for sharpness	2
45	076-EL-P009	Rechargeable NiMH battery, from kit	1
46	Nov6-foot	Gripper feet with slot for machined aluminum blades, 3D printed PLA	2
47	076-EL-P001	ESC for brushed motors, from kit	1

Table 3.3: Bill of Materials (Items 31–47)

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
48	076-EL-P002	ESC for brushed or brushless motors, from kit	1
49	Rubber Band	Tower Hobbies supplier	1
50	076-ME-P023	Bronze flanged bushing, 0.5in long, 0.25in bore	1
51	076-ME-P024	Bronze flanged bushing, 0.75in long, 0.25in bore	1
52	076-ME-P007-2	Hub mount, 0.25in bore	3
53	Gear Plate	Connects servo cradle base and threaded rod to hub gear, 3D printed PLA	1
54	93365A120	Heat-set inserts, 4-40 $\times$ 0.135" long	3
55	93365A130	Heat-set inserts, 6-32 $\times$ 0.15" long	4
56	076-ME-P005	Spur gear, 20 tooth, 0.25in bore	2
57	87921243	Hex nuts, 4-40	5
58	5506050	Socket cap screws, 6-32 $\times$ 1/2"	10
59	5504055	Socket cap screws, 4-40 $\times$ 1/2"	8

Table 3.4: Bill of Materials (Items 48–59)

Note: While the design input for many of the fabricated parts was shared by others in the group, Oliver Shapiro, Peter Phipps, and Jillian Martin shared sole responsibility for their production.

3.2 Drawings

An updated set of drawings can be found in the **appendix**.

3.3 Absolute Mass and Center of Mass

The estimated mass of our bot in Project Assignment 3 was 8.62 lbs, without a battery. The measured mass of our bot, after a significant redesign, was 4.7 lbs on competition day with the battery. The following are the center of masses, in mm, that we calculated on our final design both in its initial upright position and its position of full extension.

- **Stow Position** — $(-8.21, -13.94, 290.17)$
- **Deploy Position** — $(-136.97, 214.09, 65.80)$

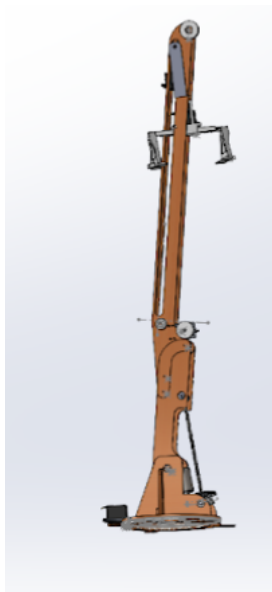


Figure 3.1: The bot in Stow Position

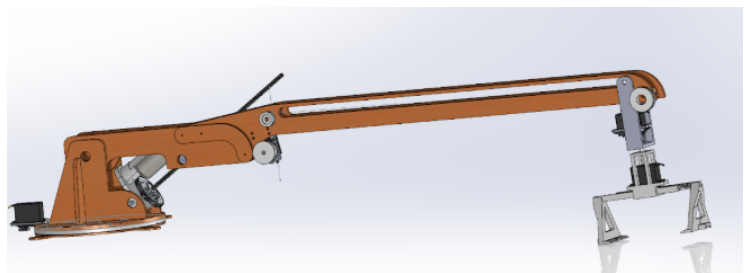


Figure 3.2: The bot in deploy position

Implementation

Given the availability of a laser cutter for manufacturing precise, lightweight plywood parts, we relied heavily on laser cut wood for the final design. All other parts were 3D printed, machined, or provided in the project kit materials.

Body

Fabricated using a large 3D format 3D printer using a combination of 3mm and 6mm Birch Plywood. Grain directions were optimized to maximize strength by analyzing the force that would act on each member.

Threaded Linear Actuator (See Figure A.5)

This was made with a three step assembly including a motor holder, 1/4-20 threaded rod, mill machined custom nut made of aluminum, gear plate fabricated through 3D printing, 72 tooth 32 pitch gear and a 10 tooth 32 pitch gear. This sub assembly was very tedious to install but once it was it was very secure, since this design only needs to be assembled upon original manufacturing. Any failure of the DC motor would have required a significant, tedious reassembly. However, this module proved to be one of the most unique and effective assemblies in the entire robot.

Rotational Base

Fabrication used laser cutting, 3D printing, aluminum stock, and glue. A base which was able to be bolted to the table was laser cut, then a lasercut insert was placed on top of that and glued down, then a bearing-restraint was placed on top, free to move radially, with 7 aluminum stock bearings placed inside, this was all sandwiched with another piece of laser cut wood which maintained the assembly.

Gripper

The gripper assembly was primarily 3D printed with one machined part. The gripper housing contains complex geometry and should minimize mass, so 3D printing was an obvious choice. The servo sits inside the housing and is secured in place using 4 heat-set inserts. The grippers arms and feet were originally designed and printed in one piece, but after printing, it became clear that the part needed to be manufactured in multiple pieces. The original one-part print failed as the entire rack needed to have supports, and the print orientation made it easy for the part to snap where the rack attached to the foot. The improved design separated the rack from the feet and allowed them to be fastened together using screws that screw into heat-set inserts. This also allowed us to interchange and iterate our foot design without reprinting our rack. Our final design for the feet also included aluminum tabs that were designed to be able to get under the ramp and effectively clamp onto the cones. The use of metal in this part allowed us to sharpen them down to a really fine edge to optimize their performance. These foot tabs were cut from an aluminum bar, tapped with threaded holes to secure them down, and then one end was sanded down to a fine edge. Screws were fastened through the top of the feet and into these tabs to secure them in place.



Figure 4.1: Gripper on the bot

Pulley System For Lateral Movement

Pulleys were fabricated using the lathe and 3D printing. The pulley itself used a thin string and was secured with 4-40 screws. After struggling to get enough friction on the drive pulley to consistently move our gripper cart, we decided to superglue a section of rubber band to the inside of the pulley in order to increase friction between the string and the pulley. This design change was ultimately a success and demonstrated how material properties like friction are very important to consider but are easily overlooked when designing and manufacturing.

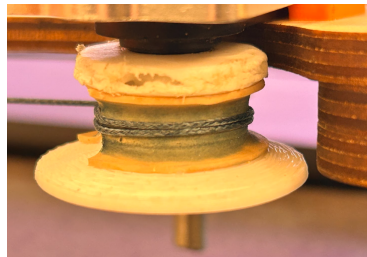


Figure 4.2: Our Rubberized pulleywheel

Cable Management (See A.6) - The cable management of our robot was constructed around the robot design after full fabrication, assembly and connection to all devices. We configured the wires while the robot was in its deploy position, to ensure that all cables could reach their designated spots. We then used different fasteners, such as zip ties and Velcro, to secure the cables in the desired locations. We also used a string running through a channel in the center of the robot arm to guide the wires and prevent tangling. The decision to ignore cable management in the design process may have saved us time leading to assembly, but overall left a less luxurious product. Had we considered

cable management in the original design, we could have better concealed cables for not only aesthetic purposes, but also resilience for the cables since they could be protected. Prior consideration would have also saved time in assembly because we would have a clear plan for cable routing, protection, and channel connection.

Performance

5.1 The Competition

The machine was able to perform well and successfully build the requisite bridge. All rotational parts functioned without difficulty and with precision. The gripper was robust and functional, and nothing in the design prevented proper functionality. In fact, the servo and motor movements were so effective and fast that every remote input had to be tuned down at least 30% in order to allow the operator to be precise in their movements. Due to user error (generated by extensive sleep deprivation), though, our team was not able to score maximal points in the competition. User error notwithstanding, the robot fully assembled the bridge in the allotted amount of time, 8 minutes, and all of the components complied as we expected. We were able to maneuver to every object in the arena, grasp, and accurately place them where needed. Our robot was also the lightest crane-like bot, highlighting our ability to manufacture an efficient, cost effective, workable robot.

Our bot did experience critical failure, though, in between rounds of testing. We failed to affix the bot to the table while not in use, and it was bumped, causing the tower to fall on the gripper with considerable speed. This fall severed the connection between the gripper and the rotational servo—a critical functional failure that we could not fix within the short timeframe between rounds.



Figure 5.1: A built bridge!

5.2 Further Takeaways

Reliability Through Redundancy

A major takeaway from this project was the importance of carrying critical spare parts, especially for components essential to core functionality. In our case, the failure point was

the 3D-printed D-shaft that coupled the gripper rotation mechanism to a hub containing the actual gripper. After an unexpected fall between runs, the shaft snapped and because we had no backup, we were unable to repair the assembly. This single failure disabled the entire gripper system and ultimately forced our forfeit from the competition. This experience highlighted how even a small, inexpensive part can become a single point of failure in a complex machine. If we had printed a spare, we could have reinstalled it in minutes and continued competing. In a real engineering context, whether a prototype demo, a grant pitch, or a company milestone presentation, such a preventable malfunction could cost not only waste time but also credibility and resources going forward. This underscored a lesson we will all take into future projects: always identify critical components and prepare redundant backups, especially for custom fabricated or high stress parts.

Manufacturing Method/ Material Selection

One major finding from this project was that we should evaluate our material choices for critical components more thoroughly, not just manufacturing for how the bot should work, but all feasible possibilities of failure. For example, the cage that allowed for the rotational component of the gripper could have been machined on the mill out of plastic stock. Making it denser and overall more durable than the brittle 3D printed filament. Another example of this would be the gearbox spool and an earlier version of the base for our testing bot. In both cases, lack of sufficient infill was the cause for failure and breakage in critical moments. We also learnt about adjusting tolerances according to manufacturing and assembly methods. We could have prevented parts not fitting together due to issues like warping during glueing, inconsistencies in stock materials (especially wood) or imperfect 3D print with leftover support material.

Assembly of the Bot

In our first revision of the robot, the assembly of all of the subassemblies was very time consuming and difficult; ease of assembly between individually designed modules was not a design priority. There was very limited room for wrenches to screw in bolts and space for the electronic components was neglected. Placement of the bot on the playing field was a tedious process. The dowels in the lazy susan were at risk of falling out during assembly and attaching the lazy susan to the base of the crane required many dowels, bolts, and tools. This tediousness is what led to a restructuring of the bot components to improve design for assembly. The redesign better integrated the lazy susan to the base and only two tools were needed for installation. Furthermore, fewer walls remained in the construction of the tower base (2, down from 4), which allowed greater access to the base for installation without deconstruction. As such, the team could bring the crane to the playing field fully constructed rather than in parts. In essence, the overall assembly of the subassemblies was more thought out due to a reduction in part count.

Gear Ratios

There are two primary gearing assemblies in our robot: the lazy susan base, where there is a gearing between the base gear and servo gear, as well as the threaded rod linear actuator which was responsible for the robots vertical movement, and had a gearing between the shaft and motor gear. These gear trains are critical because their ratios directly control the speed and torque delivered by each motor. To determine the optimal ratios for our final bot, including in the threaded rod linear actuator, we applied standard gear ratio relationships, comparing input and output torque as well as gear tooth counts. This identifies the most efficient configuration for our performance requirements. We then selected and arranged the gears from our kit to achieve ratios that maximized torque where needed and decreased speed to increase control and precision on the game board. This was significantly easier than the worm gear setup in the previous bot.

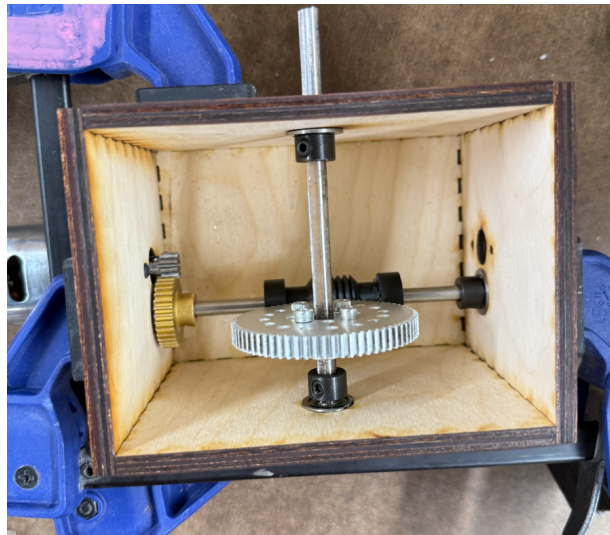


Figure 5.2: The worm gearbox from revision 1 of our project. Simple, but loud, hard to optimize, and slow in operation.

Gripping Ability

The geometrical grasping abilities of the robot's grippers were determined using solidworks. We conformed assemblies of the gripper with various parts on the game board and looked for interferences in the assembly to ensure that there were none. Then, we used free body diagrams and force analysis to assess whether the grippers would be able to pick up all of the objects on the board. After determining that this was successful, we fabricated our gripper using 3D printing and tested it with the physical components and game pieces. While the SOLIDWORKS estimations of our gripper were generally in line with our results, sometimes the computer disagreed with reality and we needed further refinements to ensure the optimization of the gripper. By following this manufacturing process, we

went through several evolutionary stages of gripper fingers, settling on a model with a continuous edge sharply tapered on the top, with a machined aluminum blade insert for specialized ramp pickup. The aluminum blades also had the added benefit of gripping the PLA cones with more consistency than the PLA teeth we had before.

Contributions

The following is a table of our general areas of technical contributions to the project:

Member	Contribution
Oliver Shapiro	Design, Fabrication, Assembly, Testing, Controls, Operator, Strategist, Troubleshooter, Design Refinement, Team Lead, Machinist
Jillian Martin	Design, Fabrication, Assembly, Testing, Report, Strategist, Troubleshooter, Design Refinement, Prototypes, Team lead, Machinist
Peter Phipps	Design, Fabrication, Assembly, Operator, Strategist, Troubleshooter, Design Refinement
Tom Thomas	Fabrication, Assembly, Prototypes, Engineering Drawings, Report, Cleanup
Lily Chabica	Prototypes, Engineering Drawings, Report, Machinist

The following table spells out the contributions made to modules in the first and second iterations of our bot, separating the stages of our redesign:

<i>Initial Design</i>						
	Vertical Motion (Worm Gearbox)	Gripper	Rotation	Lateral Motion (rack/pinion system)	Crane Arm	Base
Design	Oliver Shapiro	Peter Phipps	Lily Chabica	Oliver Shapiro	Tom Thomas	Jillian Martin, Tom Thomas
Fabrication	Oliver Shapiro	Peter Phipps	Lily Chabica	Oliver Shapiro	Tom Thomas	Jillian Martin
Testing	Tom Thomas	Peter Phipps	Lily Chabica	Oliver Shapiro	Tom Thomas	Jillian Martin
<i>Final Design</i>						
	Vertical Motion (Worm Gearbox)	Gripper	Rotation	Lateral Motion (pulley system)	Crane Arm	Base
Design	Oliver Shapiro	Peter Phipps	Lily Chabica, Oliver Shapiro	Jillian Martin, Oliver Shapiro, Peter Phipps	Oliver Shapiro	Jillian Martin, Oliver Shapiro
Fabrication	Oliver Shapiro	Peter Phipps, Oliver Shapiro	Oliver Shapiro	Oliver Shapiro, Lily Chabica	Tom Thomas, Oliver Shapiro	Jillian Martin
Testing	Oliver Shapiro, Jillian Martin	Peter Phipps, Oliver Shapiro, Jillian Martin	Jillian Martin, Oliver Shapiro	Jillian Martin, Oliver Shapiro	Oliver Shapiro, Jillian Martin	Oliver Shapiro, Jillian Martin

Reflection

This project transformed the way that we will look at and approach future design projects. While we were immensely proud of the final iteration of our bot, there was a sense of shared regret that we crammed much of the iteration and strategy adjustment in the project towards the end. The axiom of "*Start Early, Design Early, and Build Early*" is one that we often hear thrown around Thayer, but sometimes it takes a good dose of reality to reinforce the lesson.

In an ideal world, we would have traded late nights of iterations in the last few days of the project with ten weeks of thought out progression. However, if this lesson can't be taught, it is now most certainly ingrained and internalized by our group.

Regrets aside, this project was nothing but satisfying. The topics in lecture were often variations on themes that we had encountered in other classes, such as ENGS 33 (*Solid Mechanics*). After reintroduction to such topics, it was immensely rewarding to have mastery over them and be able to use them quickly and effectively in generating design analyses. For example, we were able to efficiently establish that the arm would be robust via hand calculations rather than relying on FEA or luck.

Another element that this class taught was resourcefulness. This project inherently required us to work within certain bounds. The detailed kit parts and limited materials made it so that we had to make a robot that coincided with the resources we had at our disposal. This is an important lesson on working with on hand materials which is impactful to our future in engineering sustainable and cost effective products.

Appendix

A.1 Visuals

A.1.1 Further SOLIDWORKS renderings of the our final modules

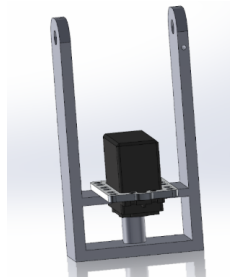


Figure A.1: Lateral Translation Cart and Rotational Gripper Assembly

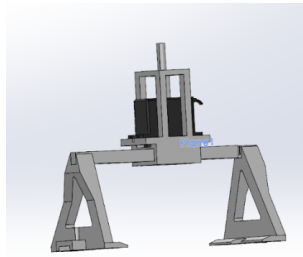


Figure A.2: Final Gripper Assembly

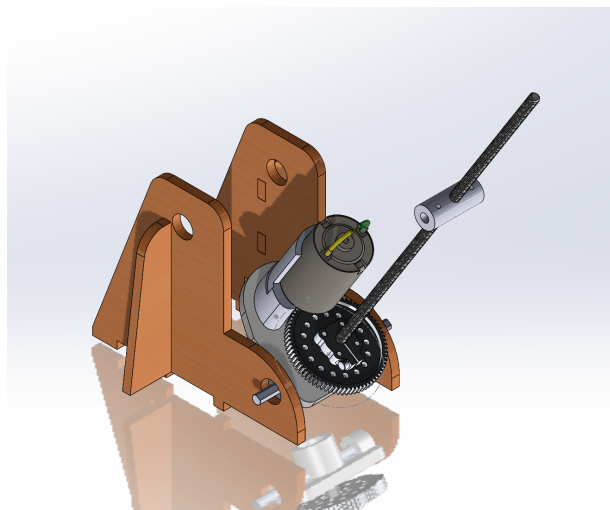


Figure A.3: Threaded Linear Actuator Module

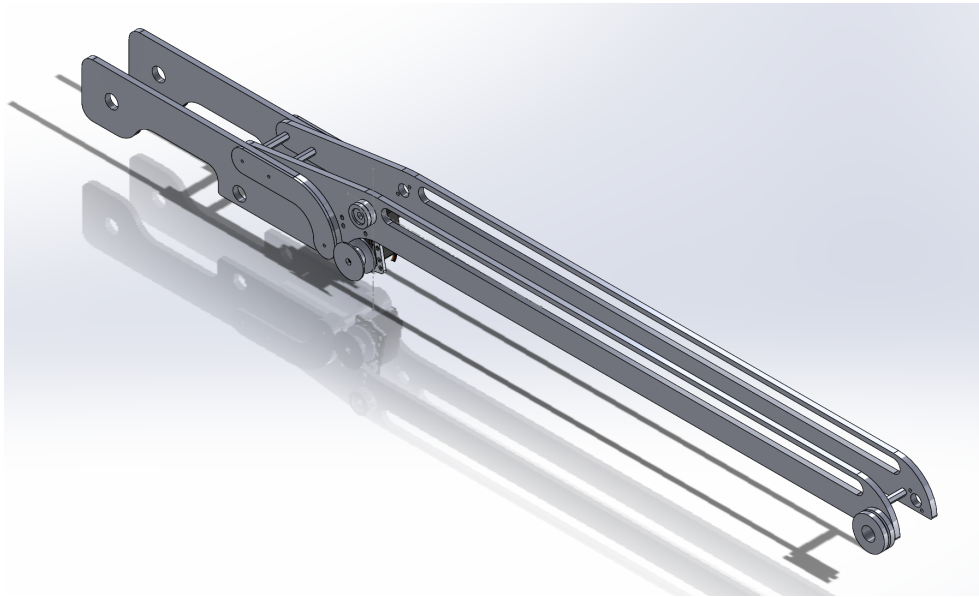


Figure A.4: Final arm design seen with pulley system

A.1.2 Photos of fabricated final modules

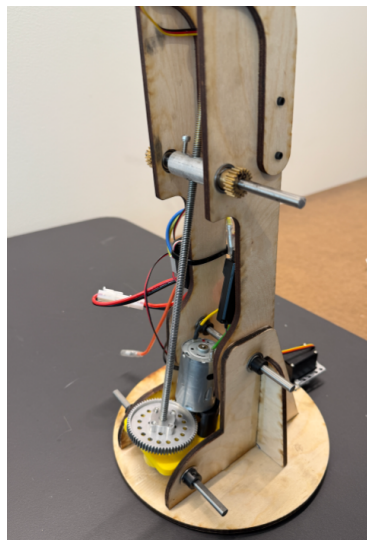


Figure A.5: The threaded linear actuator

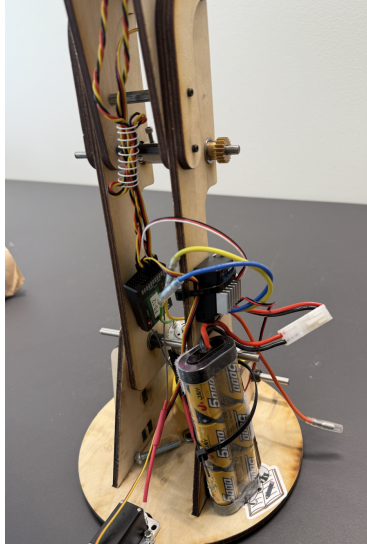
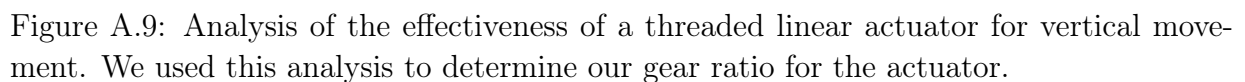
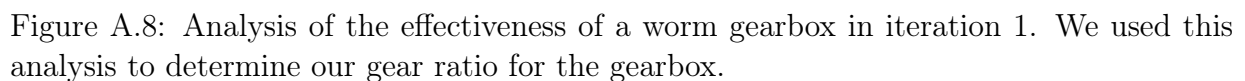


Figure A.6: Our cable management system



Figure A.7: Lazy Susan module with bearings for better rotation

The following is a non-exhaustive collection of calculations done to determine the effectiveness of various modules throughout the design process:



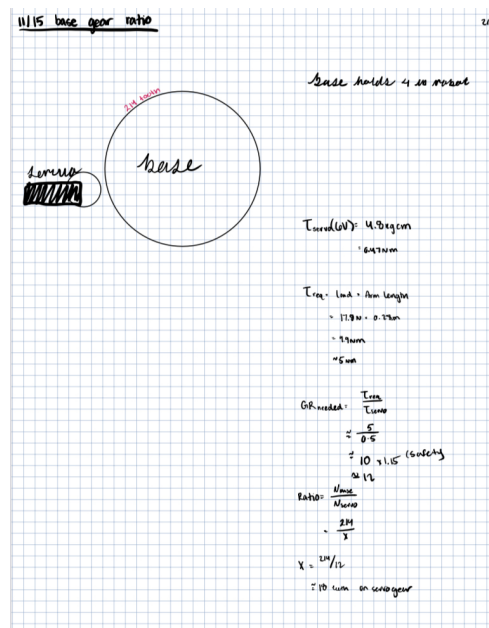


Figure A.10: Analysis for the gear ratio between servo gear and lazy-susan gear, this analysis was used to determine our gearing.

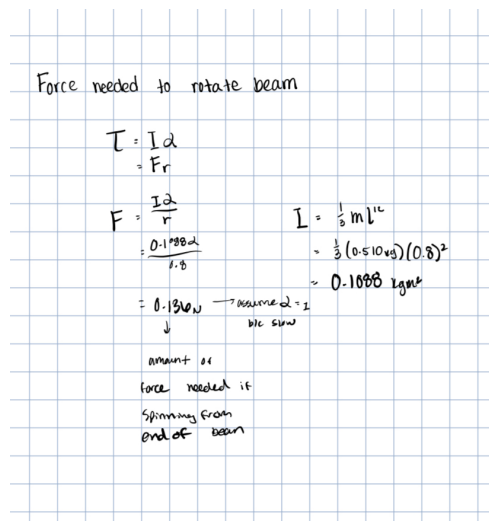


Figure A.11: Analysis for the torque needed to rotate our original revision of the robot, this was used to determine we needed a redesign.

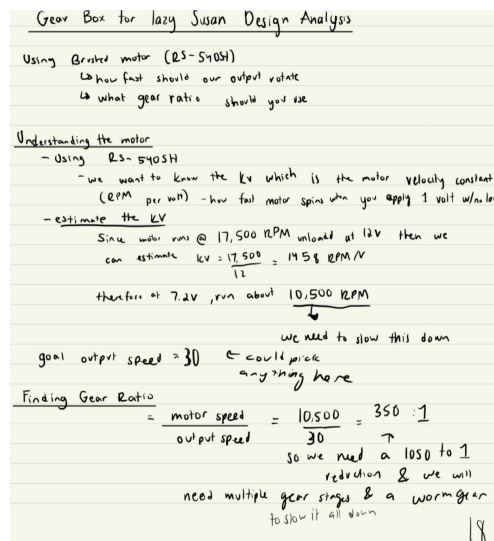


Figure A.12: An earlier lazy susan analysis

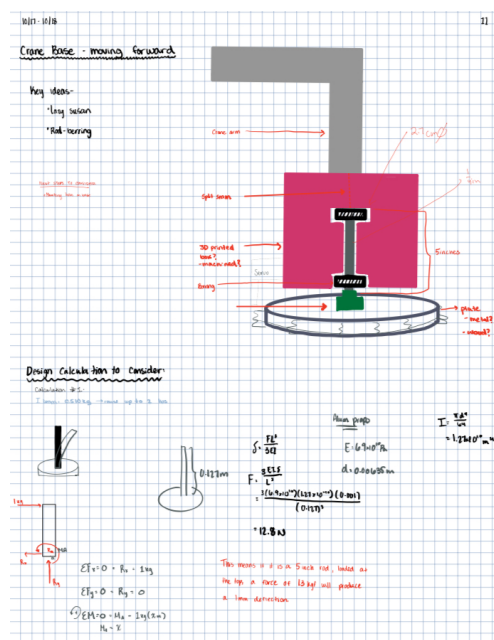
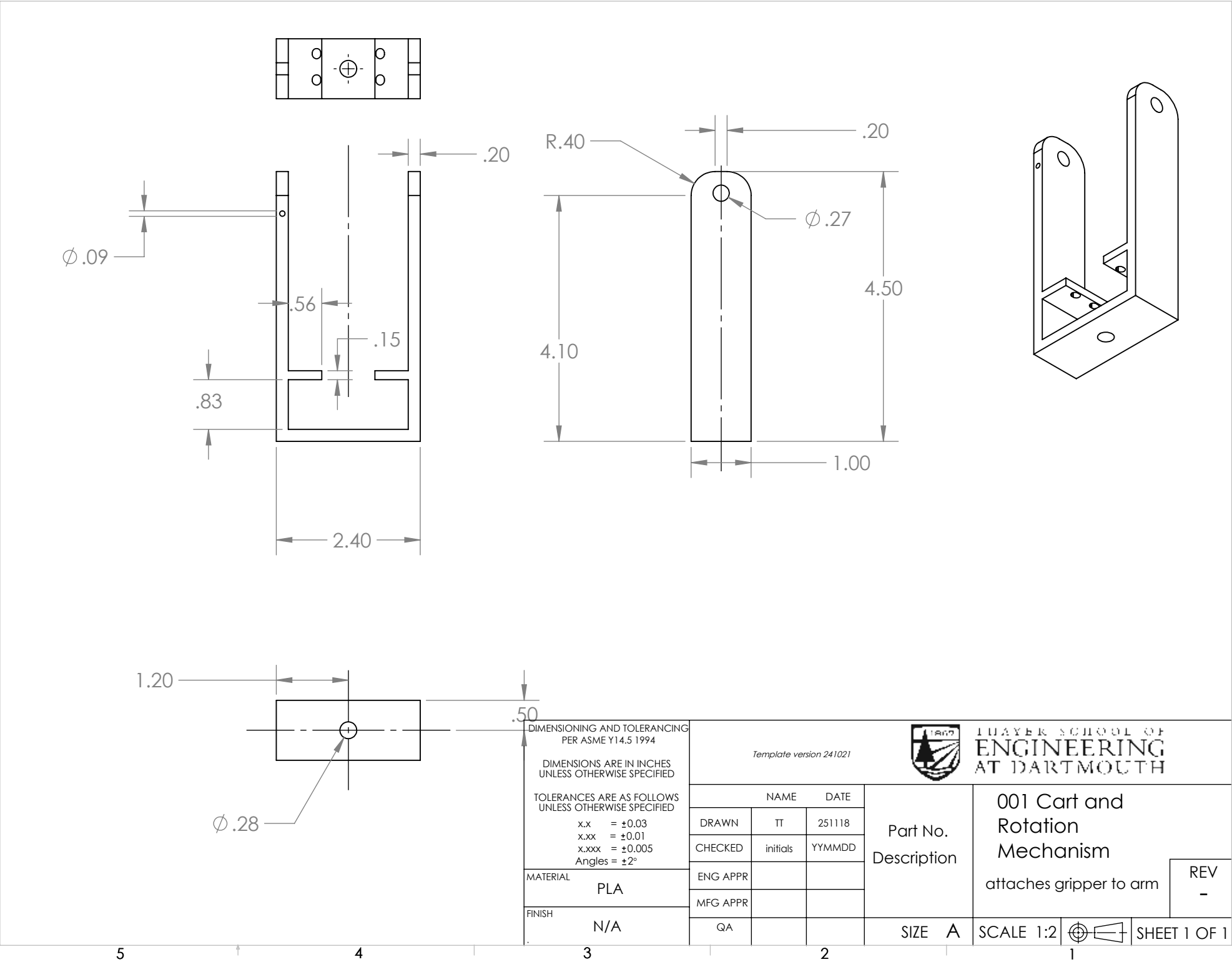
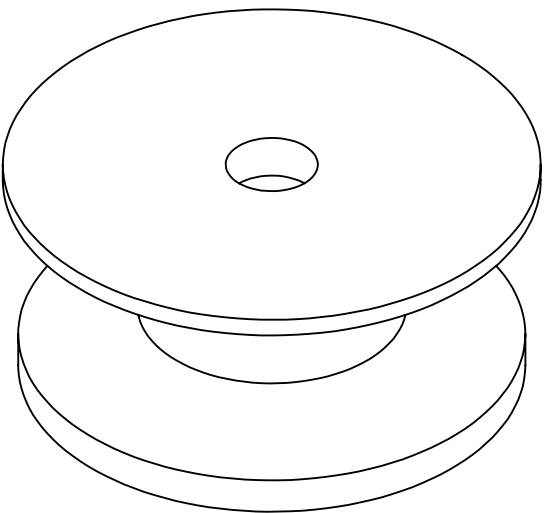
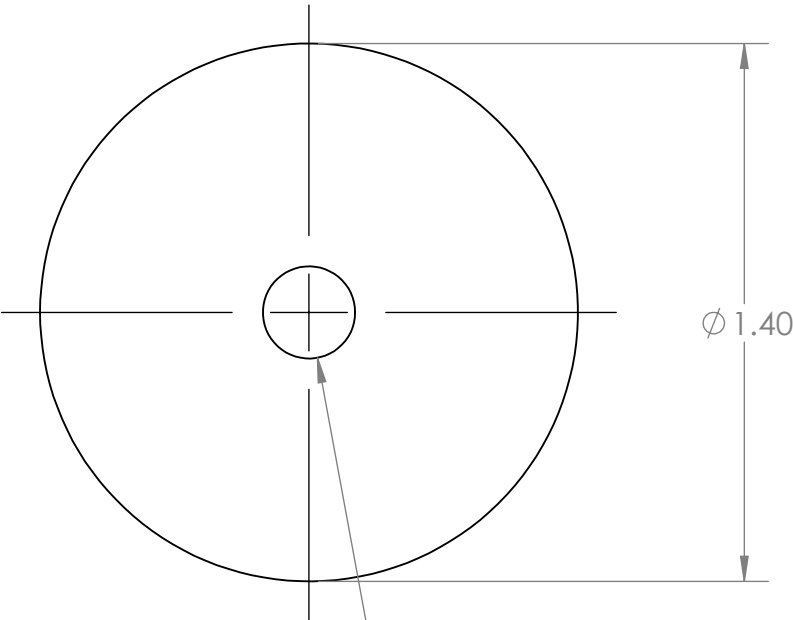
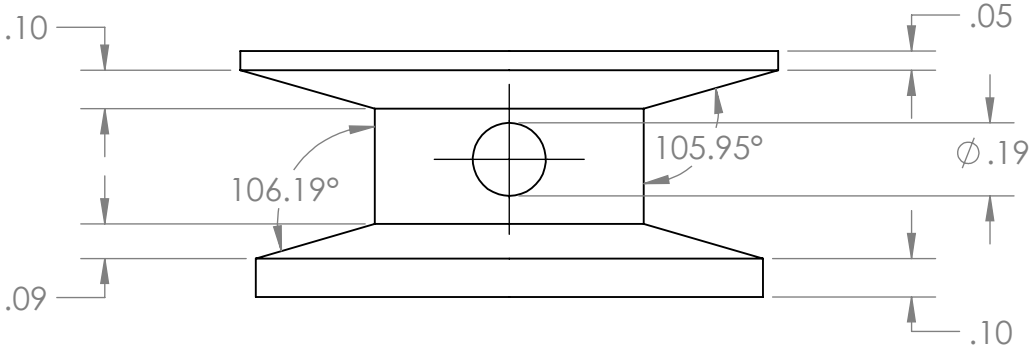


Figure A.13: Analysis of the bending happening on previous robot revisions shaft when the largest load on the end in the deploy position.





DIMENSIONING AND TOLERANCING
PER ASME Y14.5 1994

DIMENSIONS ARE IN INCHES
UNLESS OTHERWISE SPECIFIED

TOLERANCES ARE AS FOLLOWS
UNLESS OTHERWISE SPECIFIED

x.x = ± 0.03

x.xx = ± 0.01

x.xxx = ± 0.005

Angles = $\pm 2^\circ$

MATERIAL

N/A

FINISH

N/A

Template version 241021



THAYER SCHOOL OF
ENGINEERING
AT DARTMOUTH

NAME DATE

DRAWN LC 11/18/2025

CHECKED initials YYYYMMDD

ENG APPR

MFG APPR

QA

Part No.
Description

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SERVO PULLEY

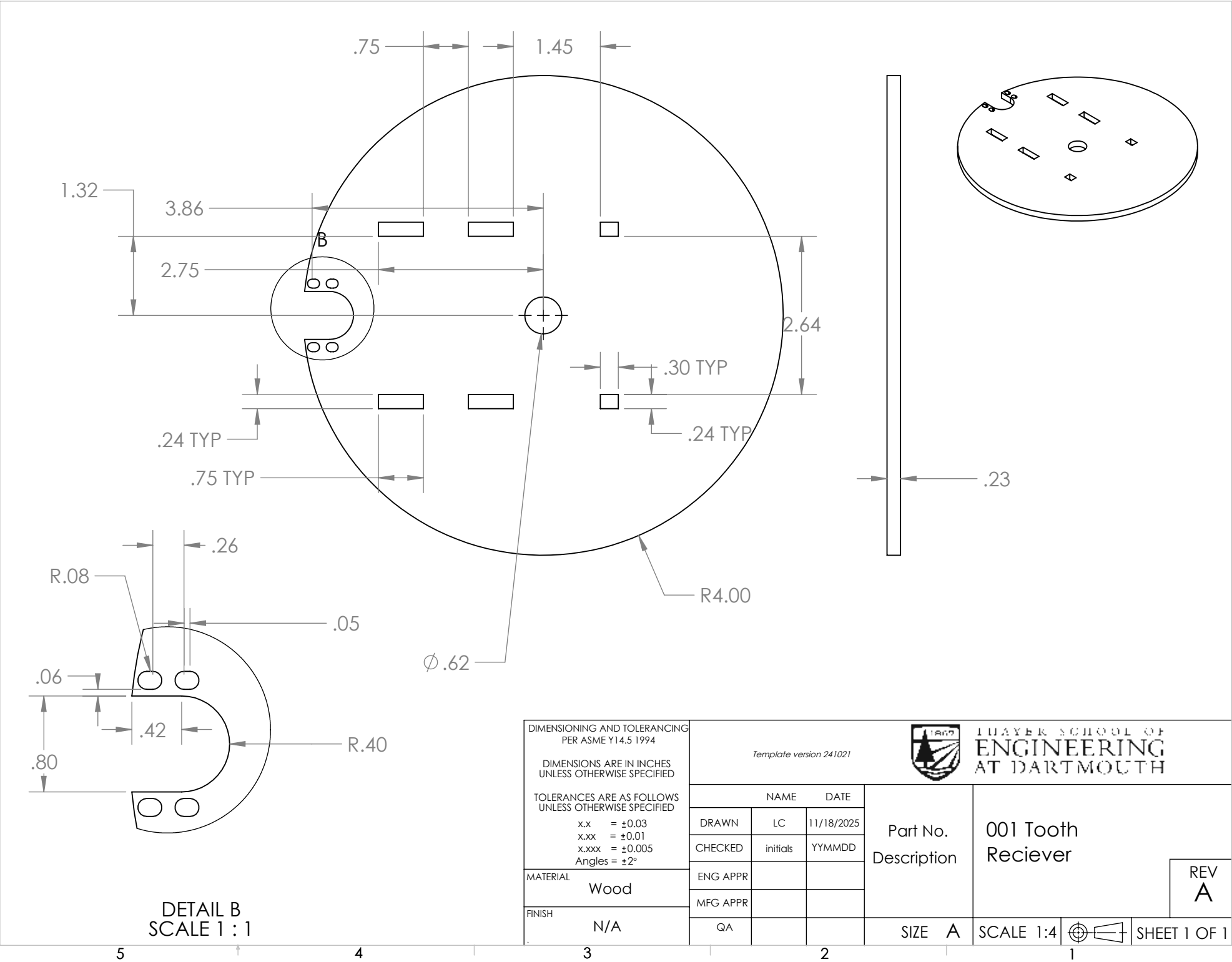
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SIZE A

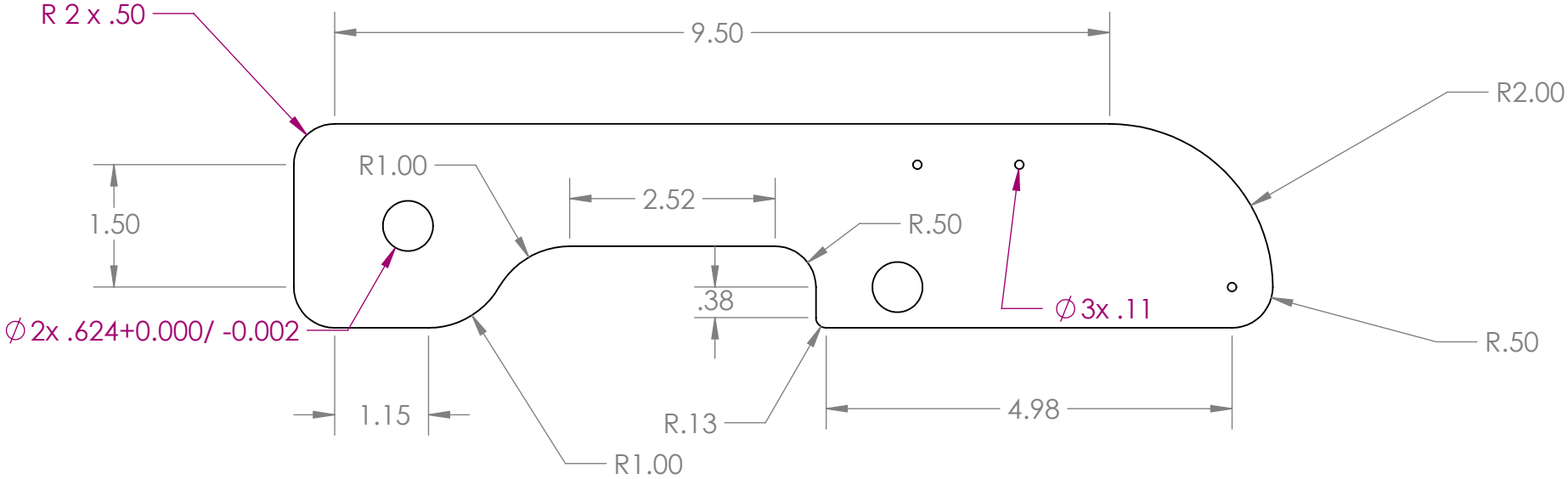
SCALE 2:1



SHEET 1 OF 1



Grain Direction



DIMENSIONING AND TOLERANCING
PER ASME Y14.5 1994

DIMENSIONS ARE IN INCHES
UNLESS OTHERWISE SPECIFIED

TOLERANCES ARE AS FOLLOWS
UNLESS OTHERWISE SPECIFIED

x.x = ± 0.03
x.xx = ± 0.01
x.xxx = ± 0.005
Angles = $\pm 2^\circ$

MATERIAL
6mm Birch Plywood

FINISH
To Cut

Template version 241021



THAYER SCHOOL OF
ENGINEERING
AT DARTMOUTH

NAME		DATE
DRAWN	TT	251118
CHECKED	initials	YYMMDD
ENG APPR		
MFG APPR		
QA		

Part No.
Description

001 Tower
Interface
short part of crane arm

REV
-

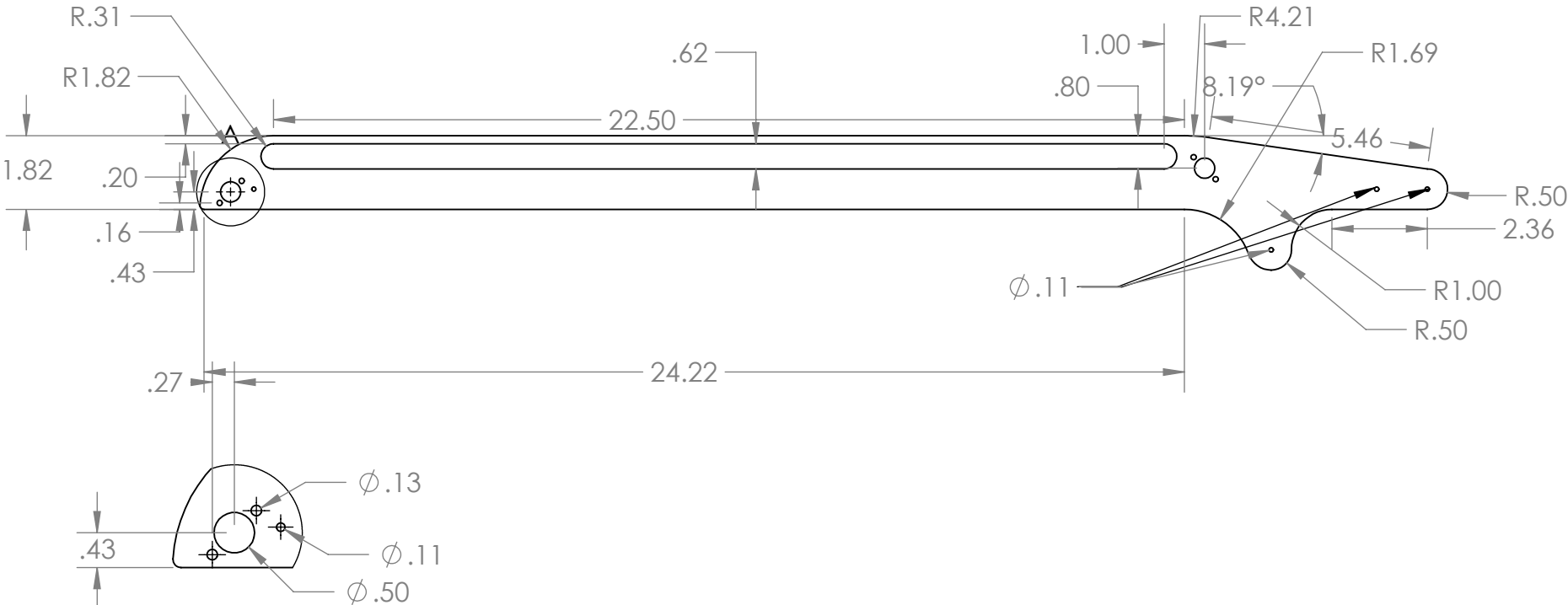
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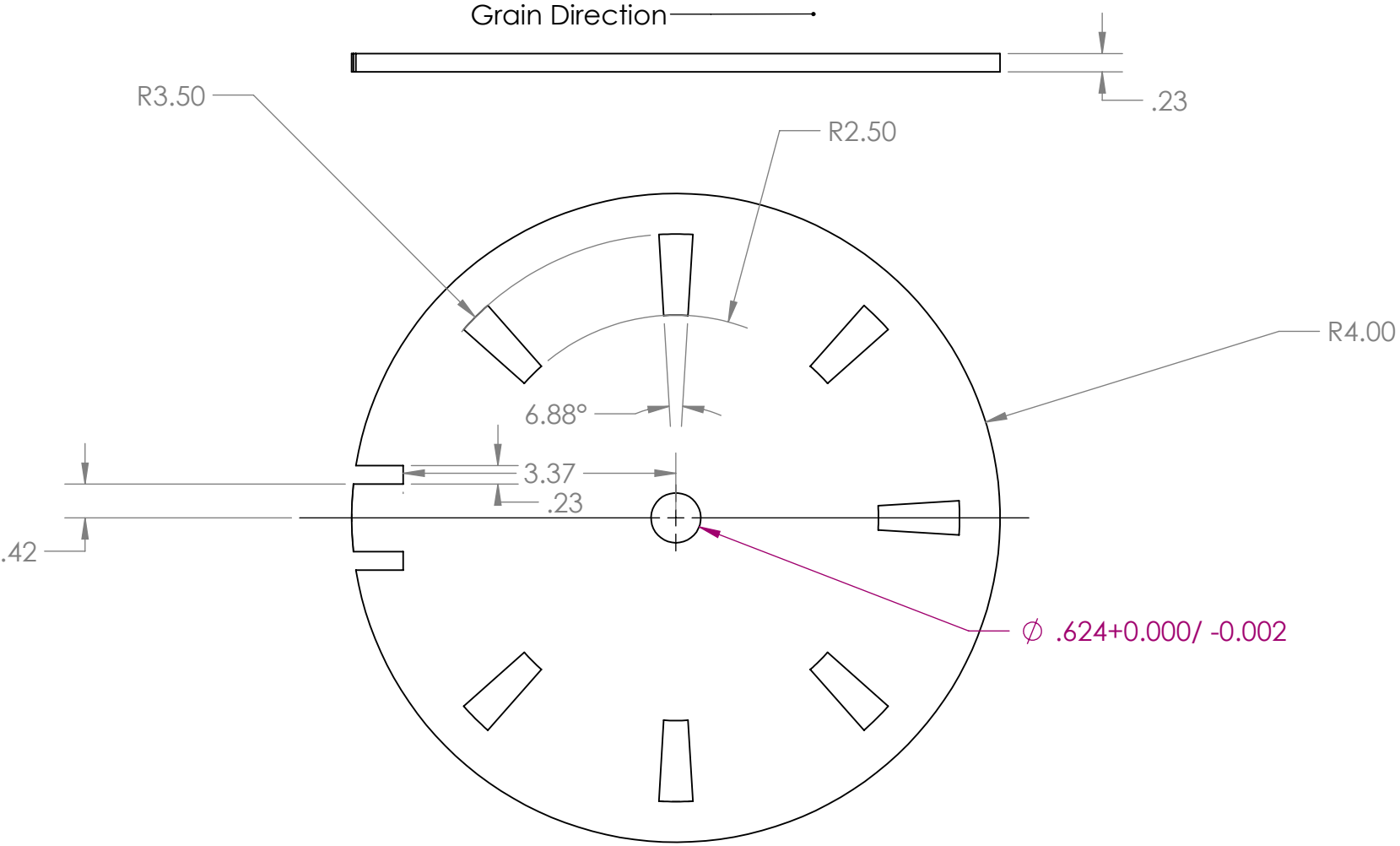
SHEET 1 OF 1

Grain direction



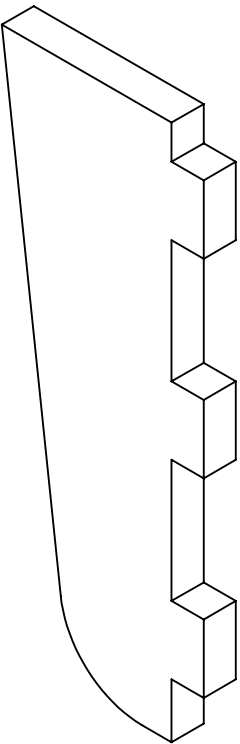
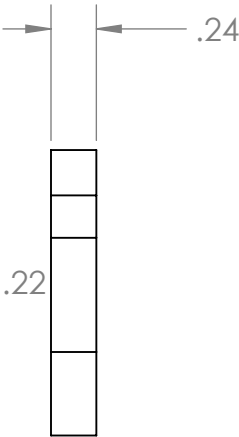
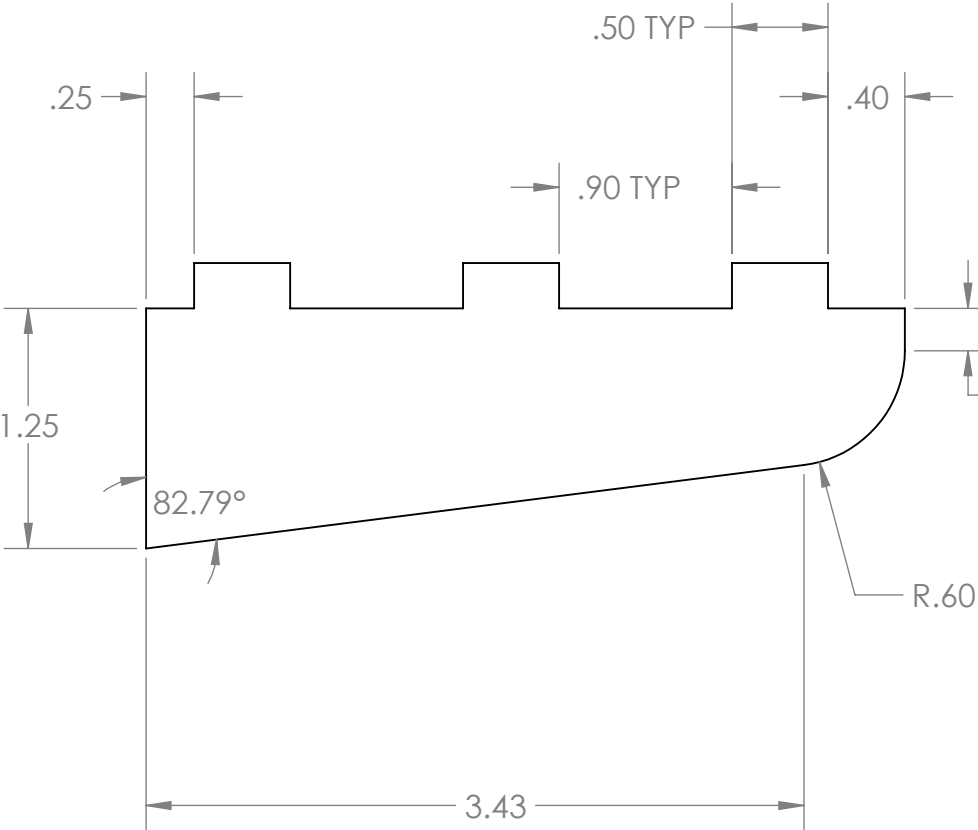
DETAIL A
SCALE 1 : 2

DIMENSIONING AND TOLERANCING PER ASME Y14.5 1994 DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED TOLERANCES ARE AS FOLLOWS UNLESS OTHERWISE SPECIFIED x.x = ±0.03 x.xx = ±0.01 x.xxx = ±0.005 Angles = ±2° MATERIAL 6mm Birch Plywood FINISH to cut	Template version 241021				THAYER SCHOOL OF ENGINEERING AT DARTMOUTH				
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	DRAWN	TT	251118						
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	ENG APPR								
	MFG APPR			SIZE A	SCALE 1:10		SHEET 1 OF 1		
	QA								



DIMENSIONING AND TOLERANCING PER ASME Y14.5 1994	
DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED	
TOLERANCES ARE AS FOLLOWS UNLESS OTHERWISE SPECIFIED	
x.x = ± 0.03	
x.xx = ± 0.01	
x.xxx = ± 0.005	
Angles = $\pm 2^\circ$	
MATERIAL	N/A
FINISH	N/A

Template version 241021			 DARTMOUTH SCHOOL OF ENGINEERING AT DARTMOUTH			
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DRAWN	Initials	YYMMDD				
CHECKED	initials	YYMMDD				
ENG APPR						
MFG APPR						
QA			SIZE A	SCALE 1:5		SHEET 1 OF 1



DIMENSIONING AND TOLERANCING
PER ASME Y14.5 1994

DIMENSIONS ARE IN INCHES
UNLESS OTHERWISE SPECIFIED

TOLERANCES ARE AS FOLLOWS
UNLESS OTHERWISE SPECIFIED

x.x = ±0.03
x.xx = ±0.01
x.xxx = ±0.005
Angles = ±2°

MATERIAL
Wood

FINISH
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Template version 241021



THAYER SCHOOL OF
ENGINEERING
AT DARTMOUTH

NAME		DATE
DRAWN	LC	11/18/2025
CHECKED	initials	YYMMDD
ENG APPR		
MFG APPR		
QA		

Part No.
Description

002 Tower Fins

REV
A

SIZE A

SCALE 1:1



SHEET 1 OF 1

5

4

3

2

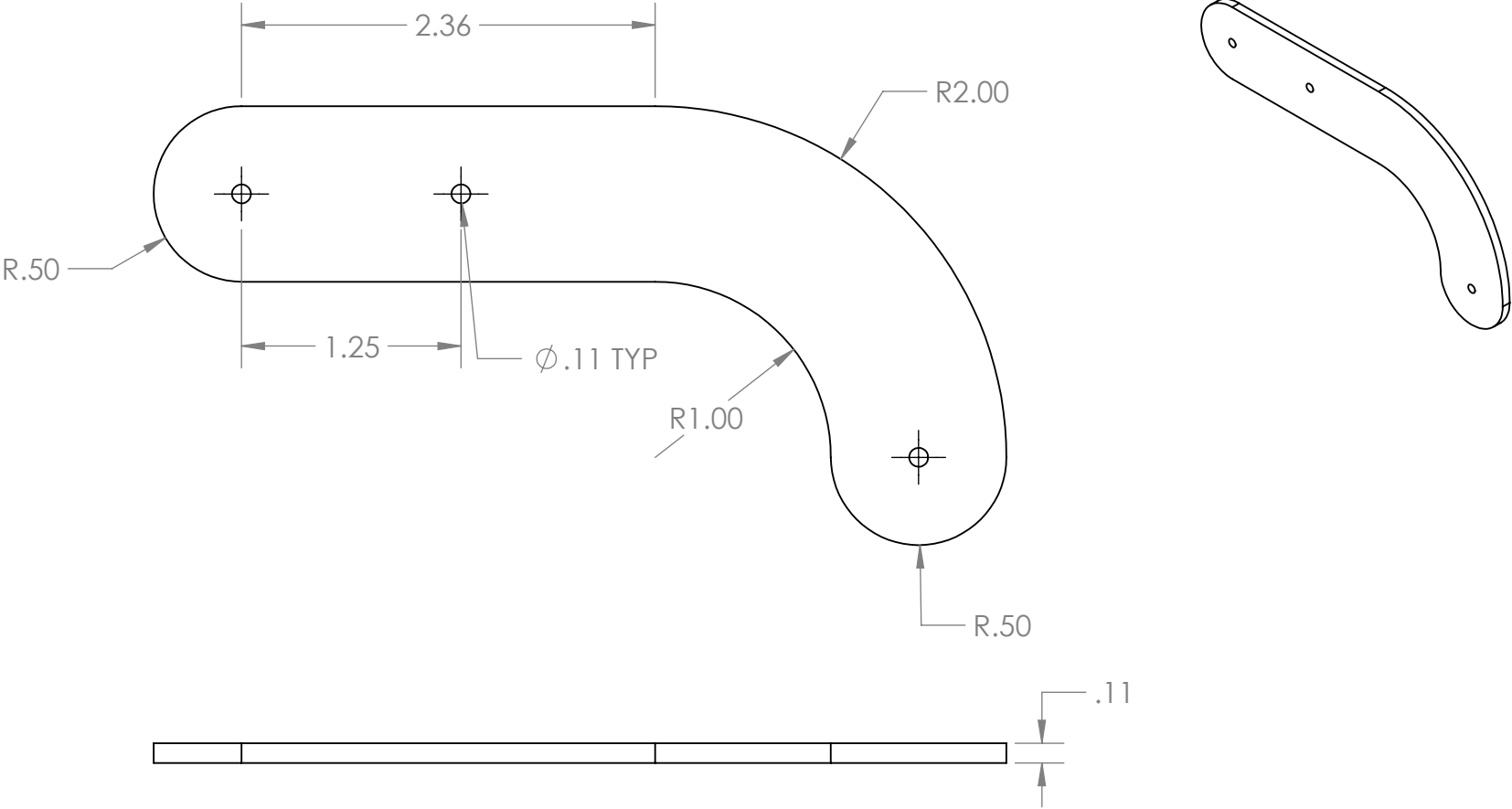
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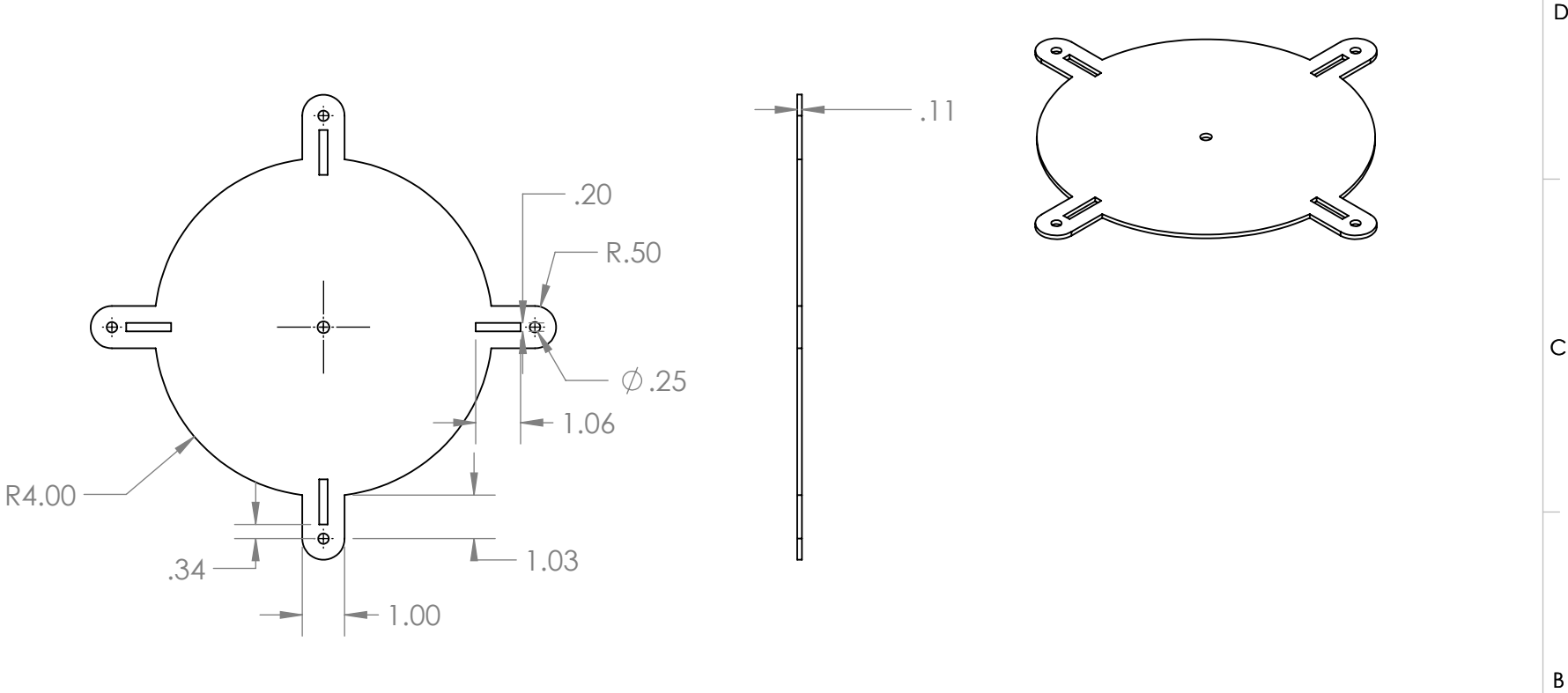
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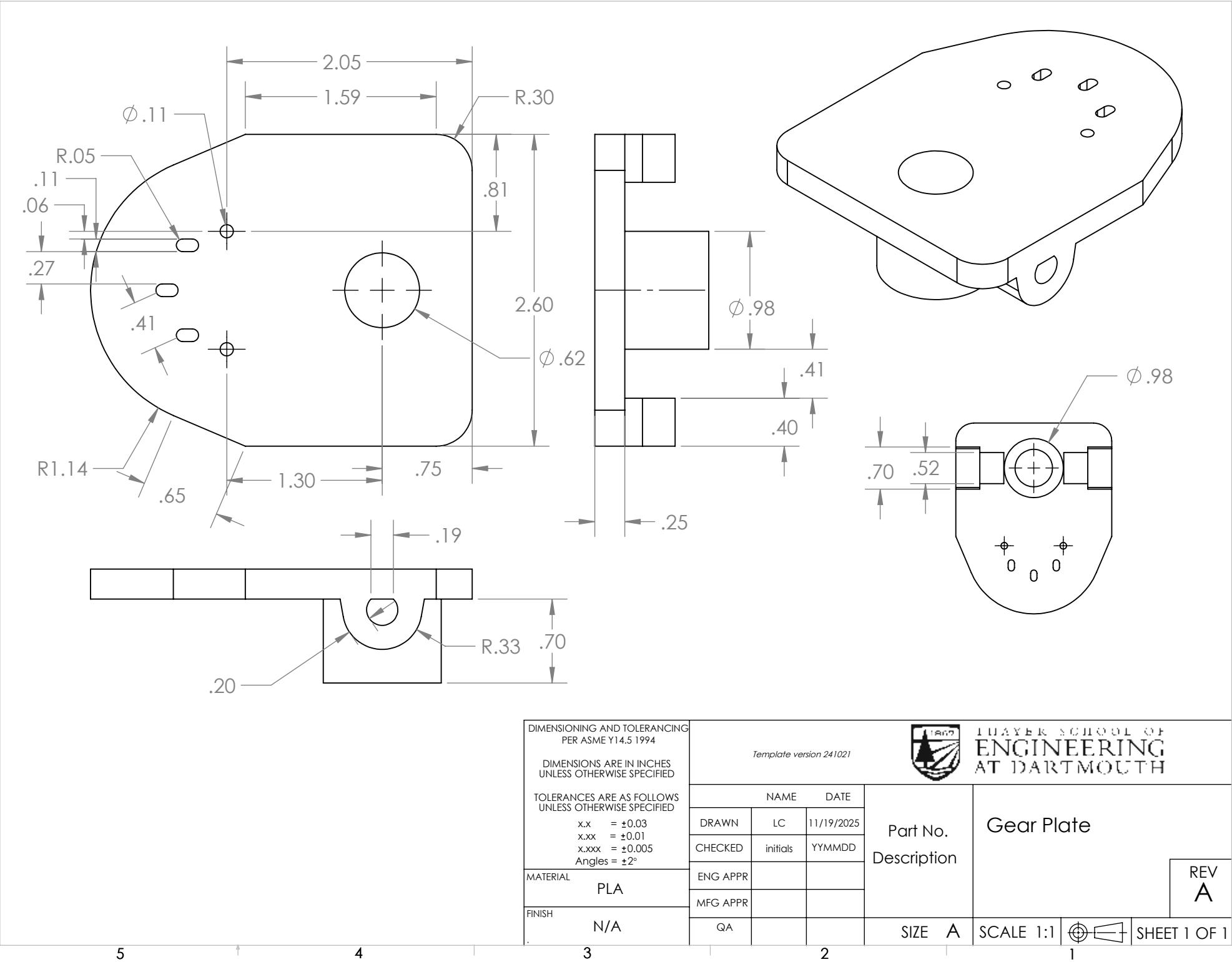
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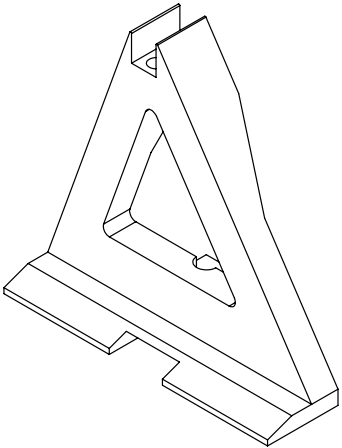
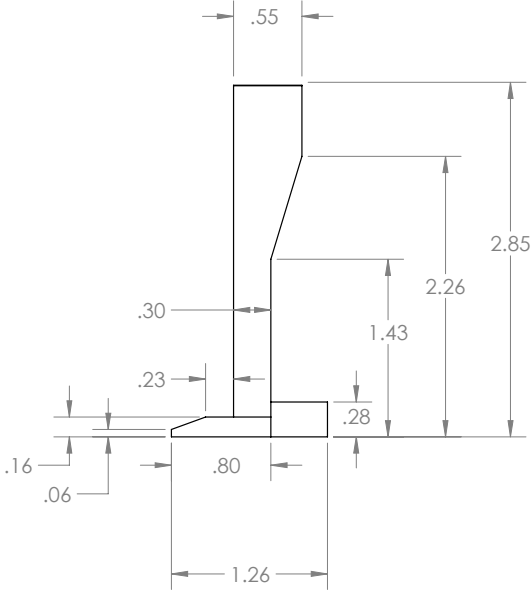
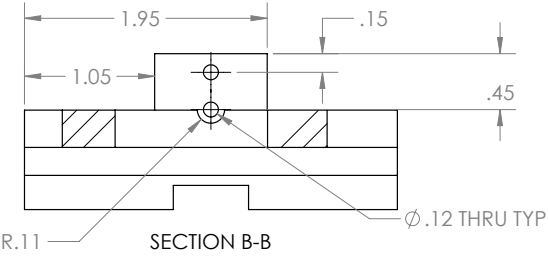
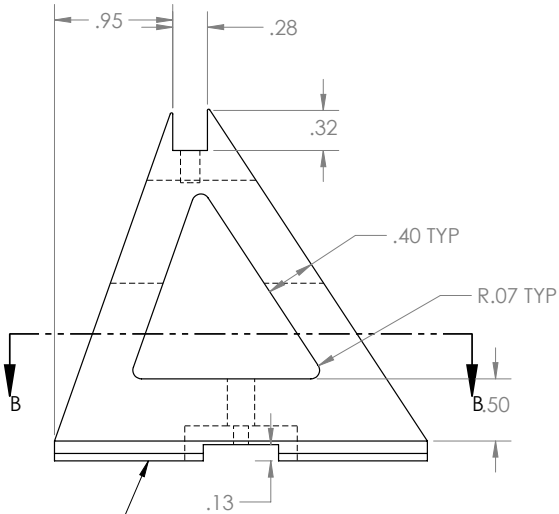
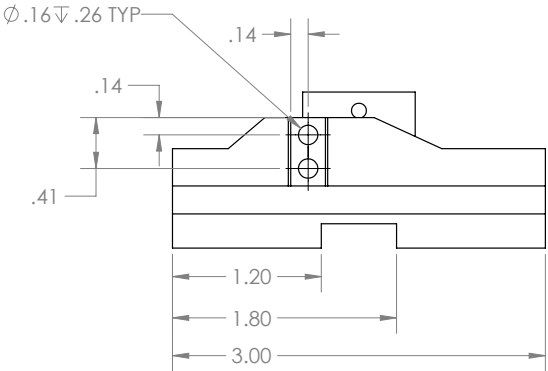


DIMENSIONING AND TOLERANCING PER ASME Y14.5 1994 DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED TOLERANCES ARE AS FOLLOWS UNLESS OTHERWISE SPECIFIED x.x = ±0.03 x.xx = ±0.01 x.xxx = ±0.005 Angles = ±2° MATERIAL N/A FINISH N/A	Template version 241021				DARTMOUTH SCHOOL OF ENGINEERING AT DARTMOUTH	
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	QA					
				SHEET 1 OF 1		REV A

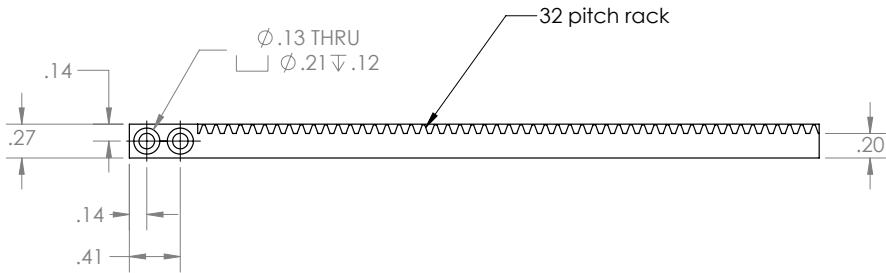
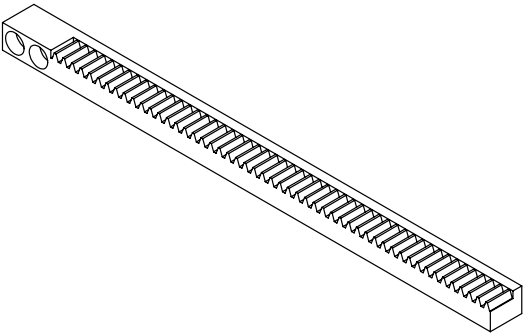
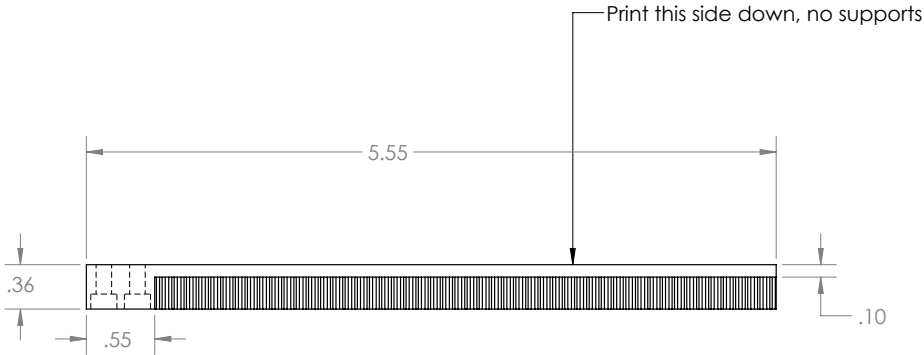


DIMENSIONING AND TOLERANCING PER ASME Y14.5 1994 DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED TOLERANCES ARE AS FOLLOWS UNLESS OTHERWISE SPECIFIED x.x = ±0.03 x.xx = ±0.01 x.xxx = ±0.005 Angles = ±2° MATERIAL Wood FINISH N/A	Template version 241021			 DARTMOUTH SCHOOL OF ENGINEERING AT DARTMOUTH	
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	CHECKED	initials	YYMMDD		
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	MFG APPR			SIZE A	SCALE 1:4
	QA				
					SHEET 1 OF 1





DIMENSIONING AND TOLERANCING PER ASME Y14.5 1994		Template version 241021		 DARTMOUTH SCHOOL OF ENGINEERING AT DARTMOUTH	
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PLA		MFG APPR			
FINISH		QA			
As Printed				SHEET 1 OF 1	



DIMENSIONING AND TOLERANCING PER ASME Y14.5 1994		Template version 241021		 DARTMOUTH SCHOOL OF ENGINEERING AT DARTMOUTH		
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